Lab 4 AVCV 2026/2027: Fire-risk classification from satellite images

CUSTOM and FINE-TUNING networks on the FireRisk subset

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1. INTRODUCTION

The goal of this lab is to assign one of seven FireRisk classes (Water, Very Low, Low, Moderate, High, Very High and Non-burnable) to a single RGB satellite image. It is an image-classification problem, not fire detection: the label describes the fire risk of the area, not whether a fire is visible. The task has two main difficulties. The classes are strongly imbalanced, from 2.4% of the images for Water to 31% for Very Low, so a model that ignores the rare classes can still look accurate; this is why the lab uses macro-F1, which weights every class equally. Moreover, the five risk levels are ordered and look alike, and neighbouring levels such as Low, Moderate and High turn out to be the main source of errors.

We work with a subset of FireRisk [1] that contains 5,861 images of 320×320 pixels. It is split in a stratified way into 3,516 training, 1,172 validation and 1,173 test images with hidden labels (class counts in Table 3). The networks are trained on 224×224 crops of these images (Section 2.1).

The CUSTOM system is a VGG-style CNN trained from scratch and refined through a staged random search; the FINE-TUNING (FT) system adapts ImageNet-pretrained torchvision models, of which Swin-T was selected. The experiments tried to answer a few concrete questions: whether the network from scratch was limited by capacity or by the data, how class weighting and augmentation affect the minority classes, and whether ensembles or retraining on train+val would help.

2. METHODOLOGY

2.1. Preprocessing and data augmentation

Images are read as floating-point RGB tensors in $[0, 1]$ and normalised with the ImageNet mean and standard deviation. The pretrained backbones require it, and we also applied it to CUSTOM so that both systems share one pipeline. During training, each 320×320 image is randomly cropped to 224×224, and every batch is then augmented on the GPU with horizontal and vertical flips, 90° rotations and random changes of gamma, brightness, contrast and saturation. The augmentation has a *light* and a *strong* preset; the search chose *strong* for CUSTOM and *light* for Swin-T. Validation images are only centre-cropped to 224×224, the setting used for model selection, while the final test predictions use the whole 320×320 image, which both networks accept because they end in global pooling. No test-time augmentation (TTA) is applied.

An aerial image has no natural "up", so flips and 90° rotations create new samples without changing the label. Colour is different, since the vegetation is probably the main visual cue for fire risk, so the colour changes were kept mild. We also tried mixup [2], CutMix [3] and TTA, but none of them improved validation macro-F1 by more than the seed noise (≈ 0.008, Section 3).

2.2. System design: CUSTOM network

The CUSTOM network (Table 5) is built only from primitive layers. It stacks four VGG-style blocks [4] of 48 to 384 channels, each with two Conv 3×3–BatchNorm [5]–ReLU layers and a 2×2 max-pooling, followed by global average pooling (GAP) [6] and a two-layer classifier with dropout, for 2.74 M parameters.

The provided reference has three blocks with a single 5×5 convolution each (6, 16 and 32 channels), no normalisation and a Flatten head. We moved away from it step by step, keeping a change only if it raised validation macro-F1. With Flatten, the first fully connected layer held about 99% of the parameters of the network, and these heads scored lower than GAP. GAP removes that layer and makes the classifier independent of where a pattern appears in the image, which suits aerial views. Next, each 5×5 convolution was replaced by two stacked 3×3 convolutions with BatchNorm. They cover the same receptive field with fewer weights and an extra non-linearity, and BatchNorm made training more stable. The network still underfitted (Section 2.4), so we finally widened it to 48 base channels (+0.03); going further (64 channels, five blocks or larger crops) gave no additional gain.

2.3. System design: FINE-TUNING network

The provided AlexNet [7] was kept as a baseline (0.452), and we compared three backbones of 24–28 M parameters, all with torchvision ImageNet-1k weights [8]: ConvNeXt-T [9], Swin-T [10] and EfficientNetV2-S [11]. EfficientNetV2-S produced a non-finite loss under fp16 mixed precision, so it was replaced by ResNet-50 [12]. In every model the 1000-class classifier was replaced by a new `Linear(in, 7)` layer (768→7 for Swin-T), and the adaptive pooling by a plain spatial mean, because adaptive pooling has no deterministic CUDA backward pass

(Section 2.4).

No layer was frozen, since satellite images look very different from ImageNet photographs and the pretrained features need to adapt to them. The inputs match the pretraining conditions (224×224, ImageNet normalisation). Because the backbone is already trained and the head is random, they use separate learning rates [13]: a small one for the backbone, so as not to destroy the pretrained features, and the same rate multiplied by 1 or 10 (chosen by the search) for the head. Biases and normalisation parameters are excluded from weight decay, and stochastic depth (drop path) [14] adds regularisation inside the backbone. Swin-T (swin_p2_1) was selected for its validation macro-F1 of 0.546; it also overfitted less than ResNet-50, with a training F1 of 0.77 against 0.99 for a similar validation score (0.539).

2.4. Training protocol and model selection

Table 1: Training configuration of the selected models.

	CUSTOM (n07_60ep)	FT (Swin-T, swin_p2_1)
Loss	Cross-entropy, class weights $\propto 1/\sqrt{n_c}$, no label smoothing, no mixup	
Optimiser & schedule	AdamW [15]; linear warm-up + cosine decay [16]; fp16 mixed precision	
Learning rate	$3.17 \cdot 10^{-4}$	backbone $2.37 \cdot 10^{-5}$, head $\times 1$
Weight decay	0.0464	0.0395
Batch / warm-up	128 / 1 epoch	32 / 2 epochs
Epochs (selected epoch)	60 (36)	30 (26)
Augmentation	strong	light, drop path 0.2
Common	Seed 42; full train split (no subsampling); Kaggle Tesla T4; PyTorch 2.11 [17]	

Table 1 lists the configuration of the two selected models. The class weights $\propto 1/\sqrt{n_c}$ are a compromise between no weighting, which lets the frequent classes dominate, and inverse-frequency weighting, which lowered macro-F1 (Section 3). The remaining hyperparameters were found with a random search [18] with median-stopping pruning [19], which stops a trial that falls behind the median of the previous ones, going from coarse to fine.

For CUSTOM, phases 1–3 explored the architecture and optimiser (24 trials) and secondary settings (16 trials), and trained three finalists for 40 epochs. All of them plateaued at 0.45–0.465 with almost equal training and validation scores; the network was underfitting, so tuning the same configuration further was unlikely to help. Phases 4–5 therefore tested targeted changes to capacity and training length (Section 3). A broader 121-trial joint search reached 0.524 on validation, but we did not adopt it because the gain did not hold on the public subset (0.47). For FT, each backbone was tried with 10 configurations for 10 epochs, and the best two of each were retrained for 30 epochs.

In every run we keep the weights of the epoch with the best validation macro-F1, and the final runs use no early stopping. Training is fully deterministic: `use_deterministic_algorithms` is enabled and the seeds of all libraries and of every DataLoader worker are fixed. Training the same epoch twice gives bit-identical weights, so the notebook reproduces the selected models on the same hardware (Tesla T4).

3. RESULTS AND ERROR ANALYSIS

Our main metric is macro-F1, the unweighted mean of the seven per-class F1 scores. Hyperparameters and checkpoints were chosen on validation macro-F1; training F1 was only a diagnostic, and the public score only external feedback, whose one influence was to discard the deep-search model (Section 2.4). Training one configuration with three seeds gave a standard deviation of about 0.008, so differences below roughly 0.01 are treated as noise.

CUSTOM. In Table 2, the reference reaches about 0.36, and our first version with a GAP head 0.440, already underfitting (training F1 0.42). Widening the network to 48 base channels with two convolutions per block raised the score to 0.496 at 40 epochs, and since the validation loss of that run was still decreasing at the end, we extended training to 60 epochs, which gave the final network (n07_60ep, 0.502).

Several other ideas did not survive validation. Inverse-frequency class weights were worse than $1/\sqrt{n_c}$ weights (last-10-epoch mean 0.431 vs 0.453), probably because they push the model too hard towards the rare classes (Water receives about 13 times the weight of Very Low). More capacity (64 channels, five blocks) and larger crops gave no gain, which is consistent with a network limited by the data rather than by its size. CutMix, 80-epoch training, ensembles and retraining on train+val did not improve on the 60-epoch model either. Per-class threshold tuning gained +0.022 on the validation set itself, but under 2-fold cross-validation only +0.005 for this network (and −0.017 for a similar one), so the apparent gain was overfitting to the validation images.

FINE-TUNING and learning curves. All modern backbones beat the provided AlexNet (0.452), and the two best (ResNet-50 0.539, Swin-T 0.546) are clearly above our best CUSTOM network, which shows how much pretraining is worth with only 3,516 labelled images. For CUSTOM, the training and validation curves in Figure 3 stay close, and validation F1 saturates at 0.49–0.50 after epoch 30: the model does not overfit, yet it

stops improving. Swin-T fits the training set much better (training F1 0.77 at the selected epoch) but improves validation by only +0.04, so most of that extra fit does not transfer.

Per-class results and confusions. In Table 3, Water (F1 0.85) and Non-burnable (0.80) are easy to recognise, while the intermediate levels are hard: Low (0.22), Moderate (0.32) and High (0.33) for CUSTOM, and the same three for FT (0.33, 0.38 and 0.41). Precision and recall are similar for all classes except Low, of which CUSTOM recovers only 19%. The confusion matrices (Figure 1) show where they go. Most errors happen among the five risk levels (77% of CUSTOM errors and 76% of FT errors), and about half of all errors fall between two adjacent levels (49% and 52%). Low is the clearest case: CUSTOM labels 40% of the Low images as Very Low, and even Swin-T sends 35% of them there. Very Low is in turn often mistaken for Non-burnable (13% and 11%).

The examples in Figure 2 suggest that part of these errors cannot be solved from the image alone. The Low image is mostly open water and both models predict Water, while the Very High image shows pale, almost bare ground that both models read as Very Low. Since the labels come from a hazard map much coarser than the image patch [1], what makes an area risky is often not visible in it. Consistently, CUSTOM reaches only 0.53 (Low 0.30) on its own training images without augmentation.

Competition feedback. On the public leaderboard (Table 4), our final submission (ID 1042782) scores 0.50 for CUSTOM and 0.49 for FT. The final CUSTOM network keeps its validation score (0.502), whereas the models chosen after the longest searches lose the most: Swin-T drops from 0.546 to 0.49, and the best model of the 121-trial search from 0.524 to 0.47. This pattern points to selection bias, since the same validation images were reused across hundreds of trials, combined with the noise of a small public subset.

4. CONCLUSIONS

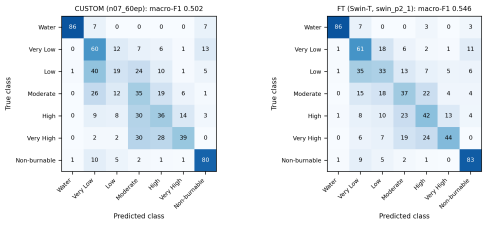
The changes with the largest supported effect were different for each system. For CUSTOM, they were those that addressed underfitting: the GAP head, the extra width and the longer schedule raised validation macro-F1 from 0.44 to 0.50, while regularisation such as CutMix did not help. For FT, the decisive factor was the pretraining itself: with a small backbone learning rate, Swin-T reached 0.546 in 30 epochs. Fine-tuning is therefore clearly better on validation, but on the public subset the two systems end up almost level (0.50 vs 0.49), so with this amount of data a well-designed network trained from scratch, with ten times fewer parameters, remains a competitive and much cheaper option.

The main limitation lies in the data rather than in the models. The intermediate classes remain hard even on training images, in line with the original FireRisk paper, which reports only 61–65% accuracy with about 91k images [1], and reusing one validation set for hundreds of decisions inflated the scores of the most searched models by up to 0.06. The next steps follow from these problems: k-fold cross-validation for a more reliable selection than a single reused split; a loss that uses the order of the classes, since most errors fall between adjacent levels; and remote-sensing pretraining and a higher input resolution, to extract more visual information.

Individual contributions.

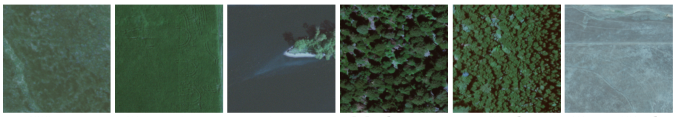
- **César Álvarez-Cascos (100475191):** design of CUSTOM search phases 1–3 (two-machine random search, finalists and the 121-trial deep search) and the GPU speed-up of training (images preloaded in RAM, batch augmentation on the GPU, mixed precision).
- **Daniel Truzman (100475161):** design of CUSTOM phases 4–5 (widening to 48×2, 60 epochs, selection of n07_60ep, deterministic Kaggle runs) and the CutMix, ensemble, threshold and train+val experiments.
- **Both:** the training runs on both members' machines, the notebook and the fine-tuning search.

Use of AI tools. We used an AI assistant as a productivity tool, always following our own criteria and instructions once we had defined the design criteria ourselves. It helped us implement and debug code, look for errors, suggest possible improvements and edit the text of this report. Every experiment, result and decision was run, checked and taken by the authors.



(a) CUSTOM **(b)** FT (Swin-T)

Figure 1: Validation confusion matrices (224 × 224 centre crop), row-normalised (%); rows: true, columns: predicted.



True	Water	Very Low	Low	Moderate	High	Very High
CUSTOM	Very Low	Non-burnable	Water	Very High	Very High	Very Low
FT	High	Non-burnable	Water	Very High	Moderate	Very Low

Figure 2: Errors shared by both models: per true class, the one with the most confident FT prediction (chosen automatically).

Table 2: Main experiments: validation macro-F1 at the best epoch (L10: mean of the last 10 epochs). Reference and selected models in bold.

Cat.	Change	Objective	Budget	Val F1
CUSTOM	**Reference notebook (StepLR)**	Baseline	60 ep	≈0.36
CUSTOM	GAP instead of Flatten (first version)	Fewer FC parameters; train F1 ≈0.42	30 ep	0.440
CUSTOM	Phases 1–3 random search	Architecture, optimiser, regularisation	15 ep (finalists 40)	0.45–0.465
CUSTOM	Inverse-frequency class weights	Help minority classes	40 ep	L10 0.431 vs 0.453
CUSTOM	48 ch, 2 convs/block (p4_09_B)	More capacity (underfitting)	40 ep	0.496 (L10 0.483)
CUSTOM	64 ch; 5 blocks; crop 288/eval 320	More capacity / resolution	40 ep	no gain
CUSTOM	**60 epochs (n07_60ep)**	Longer schedule	60 ep	**0.502** (L10 0.489)
CUSTOM	CutMix (standard; risk-only)	Regularisation	80 ep	L10 0.483; 0.470 vs 0.486
CUSTOM	Deep search (F3_B_038)	Joint search, 121 trials	80 ep, input 320	0.524
FT	**AlexNet (provided recipe)**	Baseline	5 ep	0.452
FT	ConvNeXt-T (cnx_p2_0)	Modern CNN backbone	30 ep	0.505
FT	ResNet-50 (r50_p2_0)	Residual backbone (train F1 0.99)	30 ep	0.539
FT	**Swin-T (swin_p2_1)**	Transformer backbone	30 ep	**0.546**

Table 3: Class counts and per-class validation precision / recall / F1 of the selected models (224 × 224 centre crop, as used for model selection).

Class	Train	Val	CUSTOM P/R/F1	FT P/R/F1
0 Water	86	29	.83/.86/.85	.81/.86/.83
1 Very Low	1,088	363	.59/.60/.60	.65/.61/.63
2 Low	535	178	.27/.19/.22	.33/.33/.33
3 Moderate	431	144	.29/.35/.32	.39/.37/.38
4 High	315	105	.31/.36/.33	.39/.42/.41
5 Very High	163	54	.41/.39/.40	.41/.44/.43
6 Non-burnable	898	299	.79/.80/.80	.80/.83/.81
Total / macro	3,516	1,172	.500/.508/**.502**	.539/.553/**.546**
Accuracy			.535	.577

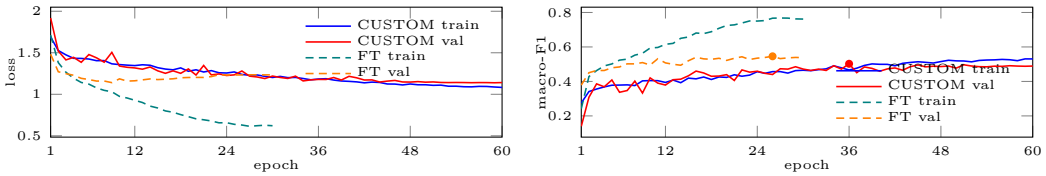
Table 4: Public CodaBench macro-F1 (fixed undisclosed test subset). Bold: final submission with the selected models; n/r: not recorded.

ID	Date	Models (custom / ft)	CUSTOM	FT
1036062	05 Nov	p4_09_B / placeholder	0.49	–
1038020	06 Nov	n07, train+val / –	0.49	–
1038186	06 Nov	F3_B_038 / F1_B_037	0.47	0.47
1038323	06 Nov	CUSTOM ens. (5) / + TTA	0.48	n/r
1042782	**08 Nov**	**n07_60ep / swin_p2_1**	**0.50**	**0.49**

Table 5: Final CUSTOM architecture.

Stage	Layers	Output
Input	RGB crop	3 × 224 × 224
Blocks 1–4	[Conv3×3-BN-ReLU]×2, MaxPool 2×2	48/96/192/384 ch
	(spatial size 112/56/28/14)	384 × 14 × 14
GAP	spatial mean	384
Head	FC 384→256, ReLU, Dropout 0.3, FC 256→7	7

2,739,447 parameters; Kaiming initialisation.



(a) Loss **(b)** Macro-F1

Figure 3: Train (with augmentation) and validation curves of the selected models; dots mark the selected epochs (CUSTOM 36, FT 26).

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